

DD_SIP-04 - Bundle Launch - Developer Implementation Guide

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1. How To Read This Guide

Bundle launch is a catalog/commercial flow.

SIP-04 creates a sellable bundle from existing SIP-01 packages, validates availability, attaches optional discount assignment rules, publishes availability, and defines allowed bundle-to-bundle migration paths. It does not migrate customers by itself.

2. Step 1 - Create Bundle Draft

Backoffice creates a bundle:

```
POST /api/commercial-bundles
Idempotency-Key: bundle-fiber-tv-starter-v1
Content-Type: application/json

{
  "operatorCode": "WIK",
  "bundleCode": "FIBER_TV_STARTER",
  "displayName": "Fiber + TV Starter",
  "description": "100M fiber with starter IPTV package.",
  "bundleType": "ACQUISITION",
  "currencyCode": "KES",
  "launchDate": "2026-07-01"
}
```

Insert:

```
insert into commercial_bundle (
  bundle_id,
  operator_code,
  bundle_code,
  display_name,
  description,
  status,
  bundle_type,
  currency_code,
  launch_date,
  created_by_user_id,
  created_at,
  updated_at
) values (
  'bun-001',
  'WIK',
  'FIBER_TV_STARTER',
  'Fiber + TV Starter',
  '100M fiber with starter IPTV package.',
  'DRAFT',
  'ACQUISITION',
  'KES',
  '2026-07-01',
  'usr-product-001',
  now(),
  now()
);
```

3. Step 2 - Add Package Components

The user selects existing packages from SIP-01:

```
POST /api/commercial-bundles/bun-001/components
Content-Type: application/json

{
  "components": [
```

```

    {"packageRef": "pkg-fiber-100m", "componentRole": "PRIMARY", "mandatory": true, "displayOrder": 10},
    {"packageRef": "pkg-iptv-starter", "componentRole": "ADDON", "mandatory": true, "displayOrder": 20}
  ]
}

```

Before insert, call:

```

GET /api/packages/pkg-fiber-100m
GET /api/packages/pkg-iptv-starter

```

Reject if a package does not exist, is retired, or has no active/launch-ready version.

Insert:

```

insert into commercial_bundle_component (
  component_id,
  bundle_id,
  package_ref,
  package_version_id,
  component_role,
  quantity,
  mandatory,
  display_order,
  metadata_json
) values
('bcomp-001', 'bun-001', 'pkg-fiber-100m', null, 'PRIMARY', 1, true, 10, '{}'),
('bcomp-002', 'bun-001', 'pkg-iptv-starter', null, 'ADDON', 1, true, 20, '{}');

```

4. Step 3 - Configure Availability

```

POST /api/commercial-bundles/bun-001/availability
Content-Type: application/json

```

```

{
  "countryCode": "KE",
  "regionCode": "NRB",
  "franchiseId": "fr_01J_NRB_002",
  "techRegionCode": "GPON-NRB-01",
  "channelCode": "SALES_APP",
  "effectiveFrom": "2026-07-01"
}

```

Insert:

```

insert into commercial_bundle_availability (
  availability_id,
  bundle_id,
  country_code,
  region_code,
  franchise_id,
  tech_region_code,
  channel_code,
  status,
  effective_from
) values (
  'bav-001',
  'bun-001',
  'KE',
  'NRB',
  'fr_01J_NRB_002',
  'GPON-NRB-01',
  'SALES_APP',
  'ACTIVE',
  '2026-07-01'
);

```

Validate franchise through EM-01 and HomePass/tech context through RLM/ILM APIs where applicable.

5. Step 4 - Attach Optional Discount Rule

If the bundle includes a commercial benefit, attach a discount rule.

```

POST /api/commercial-bundles/bun-001/discount-rules
Content-Type: application/json

```

```

{
  "discountCode": "ACQ_FREE_INSTALL",
  "assignmentScopeType": "ORDER",

```

```

    "validityMode": "ORDER_ONLY",
    "priority": 100,
    "stackingGroupCode": "INSTALLATION"
}

```

Validate discount through PLM-CFG-04:

```
GET /api/discounts/ACQ_FREE_INSTALL
```

Insert:

```

insert into commercial_bundle_discount_rule (
    discount_rule_id,
    bundle_id,
    discount_code,
    assignment_scope_type,
    validity_mode,
    duration_days,
    priority,
    stacking_group_code,
    status
) values (
    'bdr-001',
    'bun-001',
    'ACQ_FREE_INSTALL',
    'ORDER',
    'ORDER_ONLY',
    null,
    100,
    'INSTALLATION',
    'ACTIVE'
);

```

The discount rule does not create assignment now. Assignment is created later by order/campaign flow through SIP-03.

6. Step 5 - Validate Bundle

Run:

```
POST /api/commercial-bundles/bun-001/validate
```

Implementation checks:

1. Bundle has at least one mandatory component.
2. All packages exist and are active/launch-ready.
3. Component packages support requested tech/franchise availability.
4. Discount refs are active.
5. Billing event compatibility exists for bundle components.
6. No active bundle already uses same code.

Insert check rows:

```

insert into commercial_bundle_launch_check (
    check_id,
    bundle_id,
    check_code,
    check_status,
    message,
    source_ref_json,
    checked_at
) values
('blc-001', 'bun-001', 'PACKAGE_ACTIVE', 'PASS', 'All package refs are active.', '{"packageRefs":["pkg-fiber-100m","pkg-...'],
('blc-002', 'bun-001', 'DISCOUNT_ACTIVE', 'PASS', 'Discount ACQ_FREE_INSTALL is active.', '{"discountCode":"ACQ_FREE_INSTALL'

```

If any FAIL, keep bundle in DRAFT.

7. Step 6 - Submit And Approve

Submit:

```
POST /api/commercial-bundles/bun-001/submit-review
```

Patch:

```

update commercial_bundle
set status = 'READY_FOR_REVIEW',
    updated_at = now()
where bundle_id = 'bun-001'
    and status = 'DRAFT';

```

If approval policy requires review, call EM-CFG-04. SIP-04 does not own approver steps.

```

POST /api/approvals/evaluate
Content-Type: application/json

{
  "operatorCode": "WIK",
  "moduleCode": "SIP-04",
  "actionCode": "BUNDLE_LAUNCH_APPROVAL",
  "subjectType": "COMMERCIAL_BUNDLE",
  "subjectId": "bun-001",
  "context": {
    "bundleCode": "FIBER_TV_STARTER",
    "bundleType": "ACQUISITION",
    "channelCodes": ["SALES_APP"],
    "franchiseIds": ["fr_01J_NRB_002"],
    "hasDiscountRules": true
  }
}

```

If EM-CFG-04 returns `approvalRequired = true`, start request:

```

POST /api/approvals/requests
Idempotency-Key: approval-bundle-bun-001
Content-Type: application/json

{
  "operatorCode": "WIK",
  "moduleCode": "SIP-04",
  "actionCode": "BUNDLE_LAUNCH_APPROVAL",
  "subjectType": "COMMERCIAL_BUNDLE",
  "subjectId": "bun-001",
  "policyId": "apol-bundle-launch-wik-v1",
  "callbackUrl": "/internal/commercial-bundles/bun-001/approval-outcome",
  "context": {
    "bundleCode": "FIBER_TV_STARTER",
    "launchDate": "2026-07-01"
  }
}

```

Approver approves through EM-CFG-04 task API. EM-CFG-04 then calls SIP-04 callback:

```

POST /internal/commercial-bundles/bun-001/approval-outcome
Content-Type: application/json

{
  "approvalRequestId": "apr-bun-001",
  "outcome": "APPROVED",
  "decisionComment": "Approved for Nairobi sales launch."
}

```

SIP-04 patches the bundle:

```

update commercial_bundle
set status = 'APPROVED',
    updated_at = now()
where bundle_id = 'bun-001'
    and status = 'READY_FOR_REVIEW';

```

If approval is not required, SIP-04 can apply the same status transition immediately after validation.

8. Step 7 - Activate Bundle

Scheduled worker or admin action activates on/after launch date:

```

POST /api/commercial-bundles/bun-001/activate

```

Patch:

```

update commercial_bundle
set status = 'ACTIVE',
    updated_at = now()
where bundle_id = 'bun-001'
    and status = 'APPROVED'
    and launch_date <= current_date;

```

Emit:

```
{
  "eventType": "CommercialBundleActivated",
  "bundleId": "bun-001",
  "bundleCode": "FIBER_TV_STARTER",
  "operatorCode": "WIK"
}
```

Downstream systems refresh bundle availability caches.

9. Step 8 - Sales App Lists Available Bundles

Sales app calls:

```
GET /api/commercial-bundles/available?operatorCode=WIK&channelCode=SALES_APP&franchiseId=fr_01J_NRB_002&homepassId=HP-0018
```

Backend filters:

```
select b.*
from commercial_bundle b
join commercial_bundle_availability a on a.bundle_id = b.bundle_id
where b.operator_code = 'WIK'
and b.status = 'ACTIVE'
and a.status = 'ACTIVE'
and a.channel_code = 'SALES_APP'
and a.franchise_id = 'fr_01J_NRB_002'
and a.effective_from <= date '2026-07-02'
and (a.effective_to is null or a.effective_to >= date '2026-07-02');
```

Return components and discount rule summaries. Do not return internal launch approval notes.

10. Step 9 - Order Capture Uses Bundle

When a lead/order selects the bundle, FUL-02 order payload includes:

```
{
  "bundleCode": "FIBER_TV_STARTER",
  "selectedComponents": [
    {"packageRef": "pkg-fiber-100m"},
    {"packageRef": "pkg-iptv-starter"}
  ],
  "sourceChannel": "SALES_APP",
  "franchiseId": "fr_01J_NRB_002"
}
```

FUL-02 calls SIP-04 to validate the bundle is still available. If discount rules exist, FUL/SIP-05/SIP-03 creates discount assignments at the appropriate step.

11. Step 10 - Configure Bundle Migration Rule

```
POST /api/commercial-bundles/migration-rules
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "sourceBundleId": "bun-old-001",
  "targetBundleId": "bun-001",
  "movementType": "MIGRATION",
  "allowedChannels": ["BACKOFFICE", "SELF_CARE"],
  "requiresCustomerConsent": true,
  "requiresWo": false,
  "feePolicyCode": "NO_FEE",
  "effectiveFrom": "2026-07-01"
}
```

Insert:

```
insert into commercial_bundle_migration_rule (
  migration_rule_id,
  operator_code,
  source_bundle_id,
  target_bundle_id,
  movement_type,
  allowed_channel_json,
```

```

        requires_customer_consent,
        requires_wo,
        fee_policy_code,
        effective_from,
        status
    ) values (
        'bmr-001',
        'WIK',
        'bun-old-001',
        'bun-001',
        'MIGRATION',
        ['BACKOFFICE', 'SELF_CARE'],
        true,
        false,
        'NO_FEE',
        '2026-07-01',
        'ACTIVE'
    );

```

12. Step 11 - Preview Customer Migration

Backoffice calls:

```

POST /api/commercial-bundles/migration-preview
Content-Type: application/json

{
  "operatorCode": "WIK",
  "subscriptionId": "SUB-700045",
  "sourceBundleCode": "OLD_FIBER_TV",
  "targetBundleCode": "FIBER_TV_STARTER",
  "channelCode": "BACKOFFICE"
}

```

SIP-04:

1. resolves source and target bundles;
2. checks an active migration rule;
3. checks target bundle availability;
4. returns workflow owner.

Response:

```

{
  "allowed": true,
  "movementType": "MIGRATION",
  "requiresCustomerConsent": true,
  "requiresWo": false,
  "feePolicyCode": "NO_FEE",
  "owningWorkflow": "SUB-WF-MIGRATION-01"
}

```

Backoffice then calls SUB workflow. SIP-04 does not change the subscription.

13. Drools Rule Samples

Use Drools when launch/availability/migration decisions are operator-variable.

```

package rules.commercial.bundle

rule "Bundle with discount rules requires launch approval"
when
    $b : BundleLaunchFact(hasDiscountRules == true, status == "READY_FOR_REVIEW")
    $out : BundleLaunchDecision()
then
    $out.setApprovalRequired(true);
    $out.setApprovalPolicyCode("BUNDLE_LAUNCH_APPROVAL");
end

rule "Sales app availability requires active franchise"
when
    $a : BundleAvailabilityFact(channelCode == "SALES_APP", franchiseStatus != "ACTIVE")
    $out : BundleAvailabilityDecision()
then

```

```

    $out.reject("BUNDLE_NOT_AVAILABLE_FOR_CONTEXT");
end

rule "Bundle migration uses subscription migration workflow"
when
    $m : BundleMigrationFact(movementType == "MIGRATION")
    $out : BundleMigrationDecision()
then
    $out.setOwningWorkflow("SUB-WF-MIGRATION-01");
end

```

14. Suggested Worker Commands

```

commercial-bundle:activate-due
commercial-bundle:retire-due
commercial-bundle:validation-refresh
commercial-bundle:availability-cache-refresh

```

Command	Purpose
commercial-bundle:activate-due	Activates approved bundles whose launch date has arrived.
commercial-bundle:retire-due	Retires bundles with retire date reached.
commercial-bundle:validation-refresh	Re-checks package/discount refs for approved future launches.
commercial-bundle:availability-cache-refresh	Refreshes read model for sales/order/self-care lookups.

15. Error Codes

Code	Meaning
BUNDLE_CODE_DUPLICATE	Bundle code already exists.
BUNDLE_COMPONENT_REQUIRED	No mandatory package component.
BUNDLE_PACKAGE_NOT_FOUND	Component package does not exist.
BUNDLE_PACKAGE_NOT_SELLABLE	Component package is not active/launch-ready.
BUNDLE_AVAILABILITY_CONFLICT	Availability does not match component package availability.
BUNDLE_DISCOUNT_NOT_ACTIVE	Discount rule references inactive discount.
BUNDLE_VALIDATION_FAILED	Launch validation has FAIL checks.
BUNDLE_APPROVAL_REQUIRED	Bundle cannot activate before EM-CFG-04 approval.
BUNDLE_APPROVAL_REQUEST_FAILED	EM-CFG-04 approval request failed.
BUNDLE_INVALID_STATUS_TRANSITION	Requested lifecycle transition is not allowed.
BUNDLE_MIGRATION_RULE_NOT_FOUND	No active source-to-target rule.
BUNDLE_NOT_AVAILABLE_FOR_CONTEXT	Channel/franchise/region/date cannot sell this bundle.

16. Minimum Implementation Order

1. Create bundle tables and indexes.
2. Implement package lookup adapter to SIP-01.
3. Implement discount lookup adapter to PLM-CFG-04.
4. Implement bundle create/component/availability APIs.
5. Implement validation API and launch check table writes.
6. Implement EM-CFG-04 approval evaluate/request/callback integration for launch approval.
7. Implement submit/approve/activate/suspend/retire lifecycle.
8. Implement available bundles query.
9. Implement migration rule APIs.
10. Implement migration preview API.

11. Emit bundle events.
12. Patch Backoffice bundle screens.
13. Patch Sales App bundle selection.
14. Add E2E tests.

17. Test Scenarios

17.1 Launch Bundle

- Create bundle draft.
- Add two active package components.
- Add sales app availability.
- Validate, approve, activate.
- Available bundles API returns the bundle.

17.2 Invalid Package Blocks Launch

- Add retired package as component.
- Validation creates FAIL row.
- Submit review returns 409 BUNDLE_VALIDATION_FAILED.

17.3 Bundle Discount Rule

- Attach ACQ_FREE_INSTALL.
- Available API includes discount rule summary.
- Order flow later creates SIP-03 assignment.

17.4 Migration Preview

- Active migration rule exists.
- Preview returns allowed and SUB-WF-MIGRATION-01.
- No subscription state changes inside SIP-04.

18. Developer Checklist

- Bundle uses SIP-01 package refs only.
- Bundle launch does not change subscriptions.
- Availability filters by channel/franchise/region/date.
- Active bundle composition is immutable.
- Launch approval is configured/executed in EM-CFG-04.
- Drools covers configurable launch/availability/migration decisions where needed.
- Migration preview delegates execution to SUB workflow.
- Events refresh downstream caches.